A, C, M and E respectively. If A_1, A_2, A_3, A_4 and A_5 were Assistants; C_1, C_2, C_3, C_4 were Clerks; M_1, M_2, M_3 were managers and E_1, E_2 were Executive officers and if the relation R is defined by xRy, where x is the salary given to person y, express the relation R through an ordered pair and an arrow diagram.

1.5 Functions

Among several relations that exist between two non-empty sets, some special relations are important for further exploration. Such relations are called "Functions".

Illustration 8

A company has 5 employees in different categories. If we consider their salary distribution for a month as shown by arrow diagram in Fig.1.11, we see that there is only one salary associated for every employee of the company.

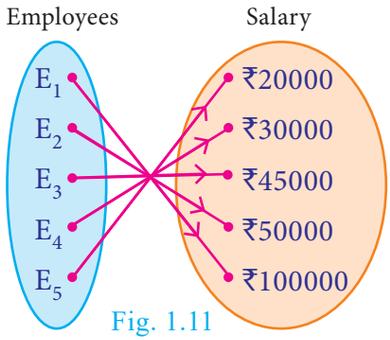


Fig. 1.11

Here are various real life situations illustrating some special relations:

1. Consider the set A of all of your classmates; corresponding to each student, there is only one age.
2. You go to a shop to buy a book. If you take out a book, there is only one price corresponding to it; it does not have two prices corresponding to it. (of course, many books may have the same price).
3. You are aware of Boyle's law. Corresponding to a given value of pressure P, there is only one value of volume V.
4. In Economics, the quantity demanded can be expressed as $Q = 360 - 4P$, where P is the price of the commodity. We see that for each value of P, there is only one value of Q. Thus the quantity demanded Q depend on the price P of the commodity.

We often come across certain relations, in which, for each element of a set A, there is only one corresponding element of a set B. Such relations are called functions. We usually use the symbol f to denote a functional relation.

Definition

A relation f between two non-empty sets X and Y is called a function from X to Y if, for each $x \in X$ there exists only one $y \in Y$ such that $(x, y) \in f$.

That is, $f = \{(x,y) | \text{ for all } x \in X, y \in Y\}$.

A function f from X to Y is written as $f : X \to Y$.

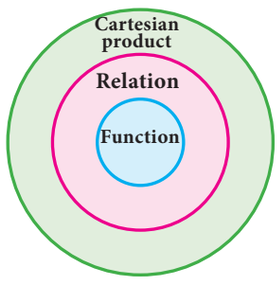


Fig. 1.12(a)

Comparing the definitions of relation and function, we see that every function is a relation. Thus, functions are subsets of relations and relations are subsets of cartesian product. (Fig.1.12(a))

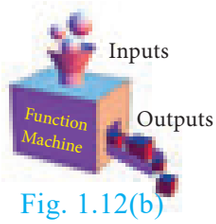


Fig. 1.12(b)

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